YANG Data Model for BGP Outbound Route Filtering

draft-zw-idr-bgp-orf-yang-model-01

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Overview

- During the review of the VPN Prefix ORF draft, the need for a YANG model was identified. The current “ietf-bgp” and its related YANG modules do not cover ORF functionality. This work extends the existing model to support ORF.
- This document defines YANG data models for the configuration and management of BGP Outbound Route Filters (ORFs), including:
  - Generic Outbound Route Filtering(ORF) capability [RFC5291]
  - Address Prefix ORF [RFC5292]
  - Covering Prefixes ORF (CP-ORF) [RFC7543]
  - VPN Prefix ORF [draft-ietf-idr-vpn-prefix-orf]
- Two YANG modules are defined:
  - ietf-bgp-orf
  - ietf-bgp-vpn-prefix-orf

ietf-bgp-orf YANG Module

```
module: ietf-bgp-orf

  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol/bgp:bgp/bgp:neighbors(peer-
groups)
            /bgp:neighbor(peer-group)/bgp:afi-safis/bgp:afi-safi
            /bgp:ipv4-unicast(ipv6-unicast):
    +--rw capability* [orf-type] {address-prefix-orf}?
    |  +--rw orf-type       identityref
    |  +--rw send-receive?   orf-send-receive
    +--rw address-prefix-orf-entries-sent {address-prefix-orf}?
    |  +--rw address-prefix-orf-entry* [sequence]
    |     +--rw sequence    uint32
    |     ...
    +--ro address-prefix-orf-entries-received {address-prefix-orf}?
      +--ro address-prefix-orf-entry* [sequence]
         +--ro sequence    uint32
         ...
```

```
augment /rt:routing/rt:control-plane-protocols
          /rt:control-plane-protocol/bgp:bgp/bgp:neighbors(peer-groups)
          /bgp:neighbor(peer-group)/bgp:afi-safis/bgp:afi-safi
          /bgp:l3vpn-ipv4-unicast(l3vpn-ipv6-unicast, l2vpn-evpn):
    +--rw capability* [orf-type] {cp-orf}?
    |  +--rw orf-type       identityref
    |  +--rw send-receive?   orf-send-receive
    +--rw cp-orf-ipv4-entries-sent {cp-orf}?
    |  +--rw cp-orf-ipv4-entry* [sequence]
    |     +--rw sequence            uint32
    |     ...
    +--ro cp-orf-ipv4-entries-received {cp-orf}?
      +--ro cp-orf-ipv4-entry* [sequence]
         +--ro sequence            uint32
         ...
```

Augments the AFI-SAFI hierarchy defined in "ietf-bgp":

- Adds ORF capability and Address Prefix ORF configuration/state
- ORF nodes are applicable to ipv4-unicast and ipv6-unicast neighbors and peer-groups
- Adds CP-ORF capability and CP-ORF configuration/state
- CP-ORF nodes are Applicable to l3vpn-ipv4-unicast, l3vpn-ipv6-unicast, and l2vpn-evpn neighbors and peer-groups

ietf-bgp-vpn-prefix-orf YANG Module

```
module: ietf-bgp-vpn-prefix-orf
  augment /rt:routing/rt:control-plane-protocols
            /rt:control-plane-protocol/bgp:bgp/bgp:neighbors(peer-groups)
            /bgp:neighbor(peer-group)/bgp:afi-safis/bgp:afi-safi
            /bgp:l3vpn-ipv4-unicast(l3vpn-ipv6-unicast, l3vpn-ipv4/6-multicast, l2vpn-vpls and
evpn):
    +--rw capability* [orf-type] {vpn-prefix-orf}?
    |  ...
    +--rw vpn-prefix-orf-quotas
    | +--rw vpn-prefix-orf-quota* [rd source-pe]
    |    +--rw rd              rt-types:route-distinguisher
    |    +--rw source-pe       union
    |    ...
    +--ro vpn-prefix-orf-entries-sent {vpn-prefix-orf}?
    | +--ro vpn-prefix-orf-entry* [sequence]
    |    +--ro sequence        uint32
    |    ...
    |    +--ro (source-pe)?
    |    |  ...
    |    +--ro route-target*          rt-types:route-target
    |    +--ro route-type?            uint8
    +--ro vpn-prefix-orf-entries-received {vpn-prefix-orf}?
      +--ro vpn-prefix-orf-entry* [sequence]
        +--ro sequence        uint32
        ...
```

Augments the AFI-SAFI hierarchy defined in “ietf-bgp”:

- Imports“ietf-bgp-orf” for ORF capability
- Adds VPN Prefix ORF configuration/state
- Applicable to l3vpn-ipv4-unicast, l3vpn-ipv6-unicast, l3vpn-ipv4-multicast, l3vpn-ipv6-multicast, l2vpn-vpls, and l2vpn-evpn neighbors and peer-groups4

Thank You